Statistics
Parameter Estimation, Hypothesis Testing, Regression

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Rules of Engagement

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- Maintain a notebook. Take notes. It helps. A lot.
- Try to revise the material at the first opportunity after the lecture. Make the opportunity.
- Learning takes time. Last minute cramming is neither efficient nor productive.
- Learning is a hard on the brain. Possibly on the body too. You have to bear the pain and make the effort.
- Exam efficiency is incidental; not the goal of the course.
- Demanding, but not cruel.

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Statistics

- Probability & Statistics is the science of uncertainty
- Probability constructs a theoretical framework, a consistent theoretical framework, derived from intuition, for statistics to be developed.
- Statistics converts observations into descriptions, rigorous and precise descriptions, about events of interest.
- The descriptions will be applicable to populations. However, studying the whole population is impractical; only a part can be examined. This part is called the *sample*.
- Methodologically, statistics studies a samples and systematically generalises the outcome to the *population* from whence the sample was obtained.
- Thus statistics is about making inferences from the sample to the population.

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Statistics

- There are some numerical facts about the population that investigators want to know, e.g., average family income in a demographic.
- These numerical facts are called *parameters.*
- These cannot be determined exactly but can be *estimated* from *samples.*
- Major Issue: Acccuracy of the estimate.
  - How close is the estimate going to be to the true value?
  - How confident are we about this closeness? i.e., what is the probability that the estimate is 'wrong.'
  - Parameters are estimated by statistics, or numbers which can be computed from the sample.
  - If $x_1, \ldots, x_n$ is the set of n measurements from n samples, some function $f(x_1, \ldots, x_n)$ is a statistic.
  - To summarise: Statistics are what the investigator **knows** from the samples. Parameters are what they **want** to know.

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Example application: Periodic Labour Force Survey

- National Sample Survey Office (NSSO) estimates of employment and unemployment indicators.
- Objective is to determine Labour Force Participation Rate (LFPR), Worker Population Ratio (WPR), and Unemployment Rate (UR).
- BTW, in Feb 2026, LFPR is 55.9, WPR is 53.2, Overall UR is 4.9, Urban UR is 6.6, and Rural UR is 4.2.
- How are these numbers obtained?
  - Determine samples of villages and urban blocks
  - First Stage Units (FSUs) are selected using *Simple Random Sampling*. 22,692 FSUs are selected in recent surveys.
  - Twelve households are selected in each FSU. 272,304 households are queried for data.
  - Convert survey into numbers for India.

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TV Advertising

- TV channel revenue is a combination of subscrition and advertisements.
- Two types of TV: Linear broadcast TV and Streaming TV
- Advertisements are most effective if they reach the intended recipient.
- Value of an advertisement to the advertiser depends on the demographic that is being targeted and the reach in that demographic.
- In streaming TV it is very easy to determine the reach in every demographic.
- It is much harder in broadcast (cable and over-the-air) linear TV.
- Television rating point (TRP) is used in determining the reach and demographic.

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Example: TV advertising

Hindi TV Serials TRP & FMN Rating

Top 5 TV Serials Of The Week

Week 10, 2026

Rank	Name	TRP	Channel
1	Kyunki Saas Bhi Kabhi Bahu Thi 2	2.0	Star Plus
2	Anupamaa	1.9	Star Plus
3	Tum Se Tum Tak	1.8	Zee TV
4	Taarak Mehta Ka Ooltah Chasmah	1.8	Sab TV
5	Vasudha	1.7	Zee TV

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TRP Ratings

TRP में फर्जीवाड़ा: रिपब्लिक विवाद के बाद ब्रॉडकास्ट काउंसिल BARC ने न्यूज चैनलों की वीकली TRP लिस्ट पर अस्थायी रोक लगाई

मुंबई 6 दिन पहले

CBI Moves To Investigate Fake Ratings Case As Complaint Is Filed In UP

Republic TV has been demanding a CBI probe, accusing the Mumbai police of going after ' after their recent clash over the Sushant Singh Rajput case.

...ported by Arvind Gunasekar, Sreenivasan Jain, Edited by Deepshikha Ghosh | Updated: October 21, 2020 7:07 am IST

TRP scam case: BARC says disappointed with Republic TV for 'disclosing; misrepresenting' confidential communication

The statement comes after the Republic Media Network earlier in the day claimed that BARC, in an email said it had found no complaint or evidence of malpractice against Republic TV, RepublicBharat or any of its affiliates

FP Staff | October 18, 2020 21:23:33 IST



TRP Ratings Scam: CBI Registers Case to Probe Alleged Fraud Based on Complaint Filed in UP

The CBI probe into the case comes amid an ongoing investigation by the Mumbai Police. Three television news channels are under the latter's radar, including the Republic TV headed ...ief Arnab Goswami.



The Manipulation of Television Rating Points: How TRPs work, the scam

Mumbai police are looking at alleged manipulation of Television Rating Points. How do channels score TRPs? What are the ways in which manipulation is possible, and how often has it been alleged in the past?

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TRP ratings: background

- TV adspend was estimated at 27,000 crore rupees in 2020. In 2025, the estimate is more than 40,000 crores
- Advertisers want to place their advertisements where they are most effective, i.e., choose slots where their target audience is present in required numbers and with required mindset to be 'programmed'.
- There are about 320 (205 rural, 115 urban) million homes in India. Out of this 230 homes have TVs. 97% of these homes are single TV homes, implying more than one viewer per TV.
- Many urban TVs are in public places (eateries) once again implying multiple viewers per TV.
- A choice of about 850 channels.
- *Audience factor* is the number of people watching per TV. This is determined by the time of day and day of week and is obtained by different methods.
- *Reach* and *Target* audience—*Gross Rating Point* and *Target Rating Point* are used by advertisers to buy ad time.

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TRP ratings: organisations involved

- In India, the popularity of different programs are measured with a number called the TRP (Televising Rating Point) rating. It is indicative of TV viewership of a program.
- "People Meters" are placed in a *sample* of the homes to determine what the households are watching.
- People meters essentially determines the channel to which the TV set is tuned. This information alongwith the times is recorded and stored suitably for later analysis.
- In India, TRP measurement is conducted by several organisations—Joint Industry Board Television Audience Measurement Research (JIB-TAM), ORG-MARG's Indian Television Audience Measurement (INTAM), (these two have now merged), Audience Measurement and Analytics (aMap), Doordarshan Audience Ratings team (DART);
- Broadcast Audience Research Council (BARC) is a cross industry body that oversees the TV audience measurement. Telecom Regulatory Authority of India (TRAI) regulates the process and provides guidelines for operation.

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TRP ratings: some methodology

- TAM is conducted in about 60,000 homes that have cable TV service by placing "people meters" in these homes.
- Channels that authorise BARC to measure their TRP rating, will include a audio watermark. People meter uses this to record the details of viewing by the family.
- Selecting the households to meter is a complex process: rural vs urban (megacities and Tier II), HSM vs South, public vs private, etc are the various criteria that will be used to select the household.
- Some years ago, households were selected from class I towns (towns with population more than 1,00,000) around the country and are supposed to be representative of the TV viewing population.
- Some obvious problems:
  - The rural population is not covered at all! Rural viewing profile is very different from urban.
  - DTH is not covered.
- Data is collected overnight and the rating published on Friday. A lot of cleaning is required.
- TRP rating determines the price of advertising in the slot.

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TRP ratings: issues

- An important question for advertiser: Do the same people watch the ad at different times?
- There is under reporting of cable homes because the service provider has to pay the broadcast company in proportion to the subscriber base.
- Channel availability is highly non uniform.
- Secrecy: The broadcasters and the cable TV suppliers should not know the families that are part of the survey.
- Effect of chasing TRP rating: Even the definition of news changes—entertainment and crime is more than 30% of the news!

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TRP ratings: some more economics

- It costs more than twenty five crore rupees per year to obtain and analyse the data annually.
- Media planning and buying agencies use these ratings to place advertising time on TV shows.
- The buyers of the analysis may pay up to ten crore rupees per month.
- More details by searching “BARC TRP Description of Methodology”
- `https://www.barcindia.co.in/measurement/television-au`

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On Sampling

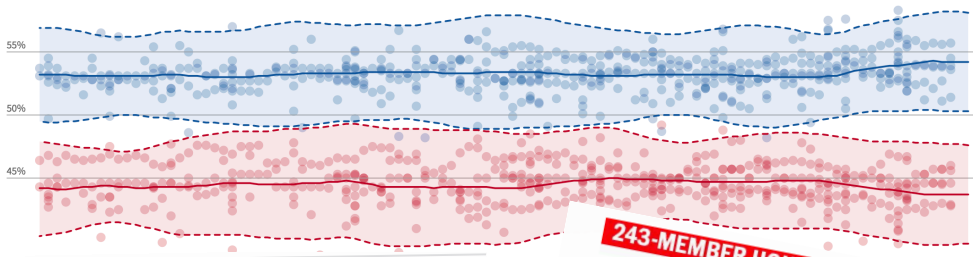
- Estimating the parameters from the sample is justified if the sample represents the population.
- This is impossible to check based on the measurements on the samples; to see whether sample looks like the population, we need to know the parameters which is what is being determined—a vicious circle.
- Thus the sampling methodology becomes important.
- The best methods involve the planned introduction of chance.
- How good can it get: In 2000, Gallup polled just 2014 people to determine that G W Bush will get 49% votes. He actually got 50.6%!
- And how bad: In the 1948 US Presidential election (Harry Truman vs Thomas Dewey) three separate polls said Dewey would get 50, 50, and 53 % of the vote and Truman would get 45, 44, and 38% of the vote. Actual result: 50% for Truman and 45% for Dewey! And here is the famous picture.

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US presidential election 2020: is Donald Trump or Joe Biden leading in our poll tracker?

Last updated: October 21

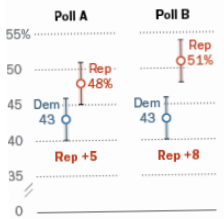
See how the opinion polls have shifted over the course of the election campaign. Each dot represents the result of a published voting intention survey while the solid lines represent our weighted polling average and the dotted lines show margin of error.



For election polls, different measures of the race have different margins of error

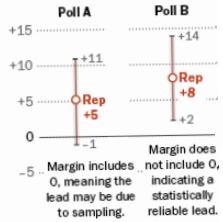
The margin of error reported for most polls applies to support for individual candidates ...

Margin of error for single candidate support (MOE +/– 3 pct. points)



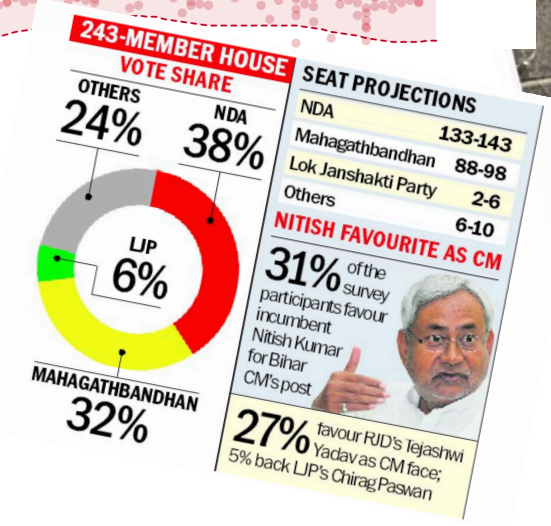
... while the margin of error for a candidate's lead is nearly twice as large.

Margin of error for difference between two candidates' level of support (%Rep – %Dem) (MOE +/– 6 pct. points)



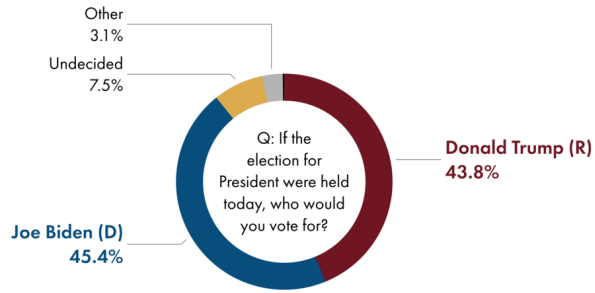
Source: Hypothetical polling results from a fictitious election.

PEW RESEARCH CENTER





Trump vs. Biden spread within the margin of error in Arizona



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A Running Experiment: Impact of MDR on UPI

- Transactions above 2000 in value at big merchants will attract an MDR of 0.4%
- What 'negative' things can happen? Make a list by listening to criticisms.
- Think of experiments to check if these things have indeed happened.
- Keep refining the questions. We will have a discussion at the end.

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Preview of the Rest of the Course

- Three key topics: Estimation, Hypothesis testing, and Regression.
- Estimation: Much like what we saw in the case studies; provide more detailed descriptions: *best* point estimates using different criteria, confidence intervals and interval estimates.
- Hypothesis testing: Make claim, formulate an statistical experiment, accept or reject the claim, and specify how sure you are about this.
- Regression: propose a relation between a set of explanatory variables and 'response variable' determine parameters of the model and/or test hypotheses about the relation.
- Key concepts to be used: pdf and pmf, Independent random variables, moments, CLT and LLN to provide some intuition.

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Recap of Probability Concepts

- Properties of probability measure
- Inclusion-exclusion, union bound and Bonferroni inequalities,
- Conditional probability and its properties: chain rule, law of total probability, and Bayes' rule,
- Independence of events
- Counting and sampling: with(out) replacement and with(out) ordering
- Random Variables and CDF
- Discrete Random Variables and PMF
- Continuous Random Variables and PDF

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Recap of Probability Concepts

- Expectation and Variance of random variables, and their properties
- Random vectors and joint CDF, independence of random variables
- Discrete Random Vectors and joint / conditional PMF
- Jointly Continuous Random Vectors and joint / conditional PDF
- Covariance and correlation coefficient
- Much of the material is covered in the text *Introduction to Probability and Statistics for Engineers and Scientists,* by Sheldon Ross.

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Gaussian Distribution

- Recall that the probability density function of a Gaussian random variable Z with mean μ and variance σ^2 is

$$f_{\mathsf{Z}}(z) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(z-\mu)^2}{2\sigma^2}}$$

- $\frac{\mathsf{Z}-\mu}{\sigma}$ has zero mean and unit variance.
- Multivariate Gaussians have a nice structure.
- Linear transformations of Gaussian random variables results in a Gaussian random variables.
- Uncorrelated Gaussian random variables are independent
- Conditional expectation of a Gaussian is linear

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Law of Large Numbers

- Let X_1 be a random variable with mean (expectation, average) μ and variance σ^2.
- Let X_i for $i = 1, \ldots, n$ be n samples of the random variable.
- Sample mean from n samples is

$$\begin{aligned} S_n &= \sum_{i=1}^{n} \mathsf{X}_n \\ M_n &= \frac{S_n}{n} \end{aligned}$$

- Important: Different realisations of n samples will result in different M_n. Thus M_n is a random variable.
- Natural question: What happens to M_n as n becomes large
- Intuitive answer: It becomes closer to μ.
- What does this mean?

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Law of Large Numbers

- As a first approximation this means that the variance of M_n decreases with increasing n.
- Recall that the variance of $\sum_{i=1}^{n} X_n$ will be $n\sigma^2$ and hence the variance of M_n would be $\frac{1}{n^2}\ n\sigma^2 \ = \ \frac{\sigma^2}{n}$.
- This implies M_n is more likely to be close to μ.
- This is the Law of Large numbers

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Central Limit Theorem

- Let us now consider a slightly different "scaling" and shifting
- Recall that $\mathsf{E}(S_n - n\mu) = 0$ and

$$s_n := \mathsf{VAR}\left(\frac{S_n - n\mu}{n\sigma}\right) = 1$$

- As n becomes large s_n 'approaches' a unit normal random variable.
- This means that for 'large-ish' n s_n can be approximated by a unit normal.

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Limit Theorems

- LLN and CLT are called limit theorems.
- They describe how random variables behave ‘in the aggregate’
- LLN tells us that the sample average gets us close to the true average
- CLT tells us how close we get to the true average
- It motivates the methods used for polling, for developing insurance schemes, for doing simulations,
- CLT justifes the widespread use of, and interest in, the Gaussian distribution.

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